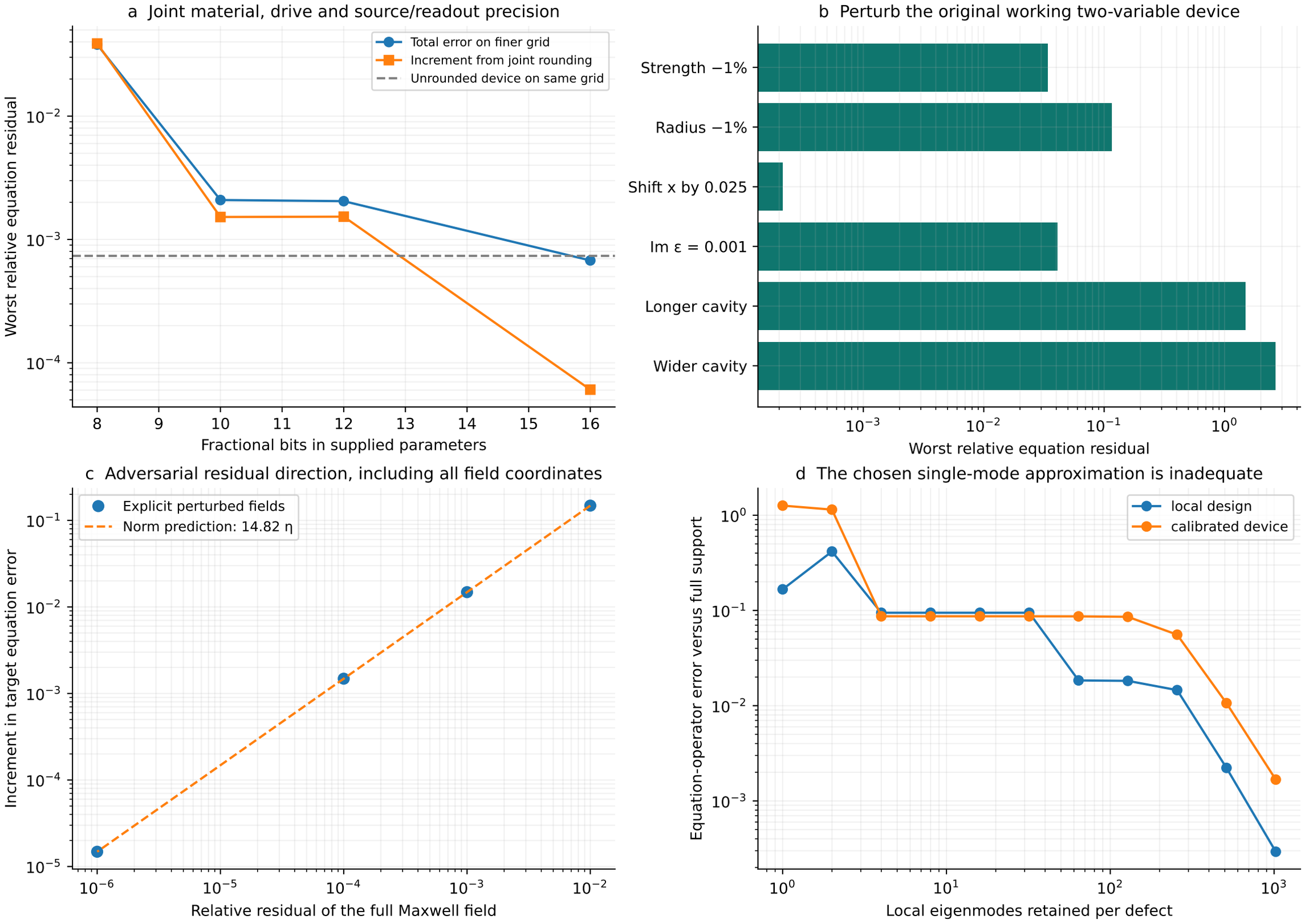
a Joint material, drive and source/readout precision
Total error on finer grid
Increment from joint rounding
Unrounded device on same grid
Worst relative equation residual
Fractional bits in supplied parameters
b Perturb the original working two-variable device
Strength −1%
Radius −1%
Shift x by 0.025
Im ε = 0.001
Longer cavity
Wider cavity
Worst relative equation residual
c Adversarial residual direction, including all field coordinates
Explicit perturbed fields
Norm prediction: 14.82 η
Increment in target equation error
Relative residual of the full Maxwell field
d The chosen single-mode approximation is inadequate
local design
calibrated device
Equation-operator error versus full support
Local eigenmodes retained per defect